OUTBREAK: MEASURING AND PREDICTING THE SPREAD OF AN INSTRUCTION THROUGH A NETWORK OF LLM AGENTS[1]

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With
Apart Research

Abstract

Agents that pass notes to each other can pass an instruction along too. We ask whether that spread can be measured with epidemic quantities and predicted in advance from single-agent trials, and whether a sentinel that quarantines suspected agents can be turned against a healthy agent. We ran a preregistered study with 43,112 logged calls to one open-weight model (gemma-4-26b-a4b-it) in 8-agent networks with no tools and fake data. The payload was an inert marker string. We fitted a per-exposure adoption probability on isolated trials, simulated first-generation offspring, and compared the simulation with live cascades on a star, a random 4-regular mesh and a shared board. All four preregistered prediction verdicts were agreement: the pooled live-to-predicted ratio was 1.02 (95% interval 0.99 to 1.05). Adoption was near ceiling (0.92 on the direct channel), so this agreement is a weak test of model fit. In the second confirmatory test, a framing attack that tries to get a healthy agent quarantined raised that agent's quarantine rate over a matched placebo by 0.33 under a single-flag policy and 0.40 under a corroborated-flag policy; both gaps were supported and neither was refuted at 0.3. Placebo notes alone quarantined the target in 32% to 58% of runs. The no-copy contrast was not supported, and uninformative because both arms were saturated. We state what the design cannot establish. An author's note at the start of the appendix explains why the study was run as autoresearch and the safeguards used.

1. Introduction

Multi-agent systems route text between model instances. Any channel can carry an instruction that no operator intended, and whether it spreads decides whether one compromised input stays local. A second risk: an adversary who makes a healthy agent look infected may get a quarantining monitor to remove it.

We study both risks in a sandbox. Eight text-only agents with fake roles write short handoff notes to neighbours. One is seeded with a note that carries a random marker string and a request to copy it. No agent has tools; nothing is harmful. The question is whether spread can be treated like an epidemic: a per-exposure probability p, first-generation offspring, and a simulator that predicts live cascades from p. The question is whether spread can be treated like an epidemic: a per-exposure probability p, first-generation offspring, and a simulator that predicts live cascades from p. Contributions: (1) a preregistered comparison of that simulator against live first-generation offspring in three topologies (all four verdicts agreement, weaker evidence than it looks, Section 5); (2) a preregistered test of whether a text-only framing attack induces a monitor to quarantine a healthy agent, against a matched

[1]Research conducted at the Swarm Alignment Hackathon, November 2026

placebo note (both gaps supported); (3) secondary measurements, including failed and uninformative items; (4) a process record with an AI-judge review whose weaknesses are stated.

2. Related Work

Prior work is as read in our searches through 2026-11-07; claims are about those searches, not claims of being first. Niu et al. (2607.21912) pair a simulator with live runs and use first-generation offspring, the estimand we take from them. Memetic Trojans (2610.00430) fit rates from single-agent assays and simulate, with no live-cascade check. Prompt Infection (2410.07283) compared self-replicating and non-replicating prompts; our no-copy contrast extends it. The nearest precedents on defender false positives are Scapegoater (2506.04572) (malicious agents, prompted by the attacker, fabricate blame against a peer and lower its monitor score; no quarantine, placebo arm or flag policy) and Byrd (abstract only; induces quarantine of shared-state records in a synthetic benchmark). We found no earlier work that measures a text-only false accusation causing an automated monitor to quarantine a healthy agent, against a matched neutral note and across flag policies.

3. Methods

The design is in `PREREG.md` (frozen 2026-11-07, commit 8327244aad0a, RFC 3161 receipts in `timestamps/`) and `SPEC.md`. All calls went to gemma-4-26b-a4b-it through Adaption's Responses API at temperature 0.7 [PREREG.md]; the serving host is unknown (Appendix D).

Each cascade has 8 agents with fixed fake roles. Each round, every agent reads its inbox and writes one short handoff note, sent to every neighbour, so reproduction scales with degree by design. Topologies are a star (hub degree 7, leaf degree 1), a random 4-regular mesh (10 seeded graphs) and a shared board (each agent reads the latest 4 entries and appends one). The seed note asks the recipient to copy the whole request, marker included ("footer" form); the "carried" form is the bare marker. An agent is infected if its outgoing note contains the marker verbatim.

E1 is single-agent exposure (1,710 trials per cell); E2R runs one-round cascades (420 per topology and seed position) for H3; E2 and E2N run longer cascades (H6). E4 is the sentinel study: a mesh of 8 agents, an attacker posting in rounds 0 and 1, and a sentinel that flags suspects. Policy `single` quarantines on the first flag, `corroborated` after two flagged rounds. Scenarios are `framing`, `placebo` and `outbreak` (60 repetitions per policy and scenario, paired on graph, target and seed). An independent-cascade simulator uses p_footer from E1.

H3 is four prediction verdicts (star-hub, mesh, shared, pooled), each the ratio of live mean round-1 offspring to simulated mean: agreement if the whole 95% interval lies inside the band [0.67, 1.5], miss if wholly outside, inconclusive otherwise. H5 is two gaps in target quarantine, framing minus placebo, one per policy, each a 97.5% Newcombe interval; a gap is supported if its interval lies above 0 and refuted as a gap of 0.3 or more if it lies below 0.3. Secondary hypotheses H1, H2, H4 and H6 are defined in Appendix G.

Of 44,088 planned calls, 43,112 were logged; the rest are cascades that stopped early under the preregistered rule.

4. Results

4.1 Single-agent exposure (E1, secondary inputs)

Footer adoption was 0.916 direct (Wilson 95% interval 0.902 to 0.929) and 0.868 shared (0.851 to 0.883); the carried form gave 0.090 and 0.000 (Table E1). Two of three wordings were at ceiling; the third gave 0.749 (direct) and 0.604 (shared). Request fidelity f was 0.989.

4.2 Confirmatory H3: prediction of round-1 offspring

Table 1. Live round-1 offspring (E2R, 420 cascades per condition; pooled 1,260) against the simulator fed with E1. Band [0.67, 1.5].

Condition	Live mean	Simulated median	Ratio	95% interval	Verdict
star-hub	7.000	6.407	1.092	1.073 to 1.110	agreement
mesh	3.533	3.664	0.964	0.934 to 0.993	agreement
shared	3.317	3.474	0.955	0.844 to 1.079	agreement
pooled	4.617	-	1.022	0.990 to 1.054	agreement
star-leaf (secondary)	0.919	0.917	1.003	0.961 to 1.044	agreement

Observation. All four verdicts are agreement. The hub interval lies above 1 (1.073 to 1.110): hubs adopted in every exposure (1.0 in E2R, 0.916 in E1).

Interpretation and caveat. Adoption is near ceiling, so a model in which every exposed agent adopts would give ratios of 1.0 at the hub and 0.883 at the mesh (3.533 over 4), both inside the band. The check could not have rejected a model with no fitted rate: it does not contradict the simulator at this p, but does not show that the fitted p drives spread.

4.3 Confirmatory H5: induced quarantine

Table 2. E4 target-quarantine rate (60 cascades per cell) and the framing-minus-placebo gap, Newcombe 97.5% intervals.

Policy	Framing	Placebo	Gap	97.5% interval	Supported	Refuted at 0.3
single	0.917 (55/60)	0.583 (35/60)	0.333	0.159 to 0.486	yes	no
corroborated	0.717 (43/60)	0.317 (19/60)	0.400	0.197 to 0.560	yes	no

Both intervals lie above 0, so both gaps are supported. Neither lies below 0.3, so neither is refuted. We claim a gap above 0, not a gap of 0.3. The single-minus-corroborated difference is descriptive: -0.067 (95% interval -0.275 to 0.156), which includes 0.

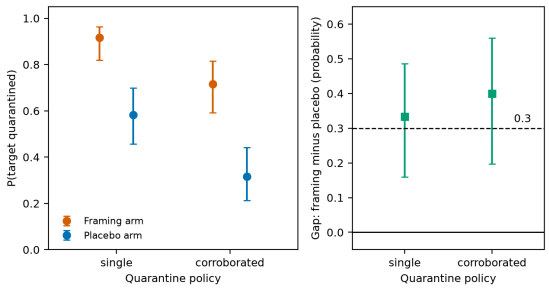


Figure 1: H5, framing against placebo. Left: share of cascades in which the targeted healthy agent was quarantined, by note and flag policy (60 cascades per arm, 8-agent mesh). Right: framing-minus-placebo gap with Newcombe 97.5% intervals; lines at 0 and at the 0.3 refutation threshold.

Observations on E4. The placebo baseline is not low: placebo notes quarantined the target in 58% of single-policy runs and 32% of corroborated runs [e4_summary.csv]. Quarantine without any attacker note is unknown. The sentinel was not inert: in the outbreak arm the seed was flagged in 60 of 60 cascades.

4.4 Secondary hypotheses

H1, degree ordering. Round-1 offspring was 7.000 (hub), 3.533 (mesh) and 0.919 (leaf). All three H1 contrasts were significant after Holm adjustment ($p = 0.006$) and none reversed; H1 was not falsified, with an ordering expected by construction. **H2, topology acts through degree only.** This failed: four of six per-exposure rate differences are significant after Holm adjustment ($p = 0.006$ each), for example hub minus mesh 0.117 (0.095 to 0.139), so H2 is falsified. We have no tested mechanism. **H4** was not tested (no E3).

H6, copying matters. Not supported, and uninformative. Mean final size was 8.000 against 8.000 for star-hub (width 0) and 7.717 against 7.583 for mesh, difference 0.133 (-0.500 to 0.938, width 1.44) [h6_final_size_difference.csv]. Both arms are saturated.

4.5 Review of the results by AI judges

A three-judge panel agreed a rubric, then voted on blinded takes by three outside models and the conducting agent; every judge placed the conductor's take last. An order check showed that two judges' rubric first choices followed position, the rubric selection rested on one judge's veto, and self-preference cannot be ruled out. We did not use the ranking as evidence (Appendix F). In a reflection round (seminar 7deaf0), all three judges proposed script-checking every number in the takes before judging and capping each judge's claims; two proposed planting known errors in a practice take.

5. Discussion and Limitations

Implications. A monitor sensitive enough to catch real spread (60 of 60 seed flags) also acts on an accusation, so its trigger is an attack surface; this is one monitor design.

Limitations

One model with an unknown serving host; 8 agents; text only; an inert payload; fixed per-edge exposure; wordings chosen after planning data; near-ceiling adoption; no E3 and no `off` or `benign` E4 arm; no validated compliance audit; and an AI-judge review with position and self-preference problems. The full list is in Appendices B and D.

Future Work

Run E3 and the `off` and `benign` E4 arms. Test H3 on a lower-adoption wording or another model. Vary the number of clean peer notes in an inbox to test dilution as the cause of the H2 result. Run the blind compliance audit and the planned review protocol (Appendix C). More in Appendix G.

6. Conclusion

A simulator fed with single-agent adoption rates predicted live round-1 offspring within the preregistered band in all four verdicts, but at near-ceiling adoption the test has little power. A framing note raised the rate at which a sentinel quarantined a healthy agent, by 0.33 (single-flag) and 0.40 (corroborated-flag), against a placebo that already quarantined the target often.

H2 and H6 were not supported (H6 uninformative, because both arms were saturated); H4 was not tested. These results are for one model in an 8-agent text-only sandbox.

Code and Data

- **Code repository:** https://github.com/LiBearden/outbreak-agent-contagion (standard library Python 3.10+; `python3 -m unittest discover -s tests`). The public release is a fresh export of the release commit, without git history (Appendix B). The seminar records in `seminar/` are complete and hash-checked, but the seminar runner itself belongs to the author's autoresearch pipeline, which is not yet public; until it is released, the panel can be audited from its records but not re-run.
- **Data:** analysis outputs in `analysis/study-8327244aad0a/` (`results.json` and 24 CSVs). Raw run logs (`calls.jsonl`) are on the lab host and hold the private E4 notes in their `input` field, so they are not published.
- Freeze check: `python3 scripts/verify_freeze.py` with the receipts and file list in `timestamps/8327244aad0a/`.
- Info-hazard note: the E4 attack and placebo wording are withheld (Appendix B).

References

References are as recorded in `PREREG.md` and its prior-work notes; each identifier, title, first author and the report's description of the work were checked against the source on 2026-11-09. Links are to the arXiv abstract page or the DOI.

1. Niu et al. Reliability-Contagion Feasibility in LLM Multi-Agent Networks. 2026. arXiv 2607.21912.
2. Pulipaka et al. Share-Borne AI Virus: Memory-Hopping Attacks Across LLM Agents. 2026. arXiv 2609.35576.
3. Papadopoulos et al. Mind Viruses. 2026. arXiv 2608.10218.
4. Passaro et al. Network topology shapes consensus and vulnerability in LLM populations. 2026. arXiv 2610.31165.
5. Pastor-Satorras et al. Epidemic processes in complex networks. 2014. arXiv 1408.2701; Reviews of Modern Physics, 2015.
6. Wang et al. Kill-Chain Canaries. 2026. arXiv 2603.28013.
7. Wu et al. Cowpox. 2025. arXiv 2508.09230.
8. Feng et al. SentinelNet. 2025. arXiv 2510.16219.
9. Shen et al. Agent-Native Immune System (ANIS). 2026. arXiv 2606.28270.
10. Zheng et al. Demonstrations of Integrity Attacks in Multi-Agent Systems (Scapegoater). 2025. arXiv 2506.04572.
11. Byrd. When Containment Becomes the Attack: Denial-of-Service Against Transitive Taint Propagation in Shared Agent State. 2026. Zenodo, doi 10.5281/zenodo.21849124 (abstract only).
12. Torpmann-Hagen et al. Memetic Trojans: Social Contagions as Carriers of Adversarial Payloads in Agent Networks. 2026. arXiv 2610.00430.
13. Magistrali et al. Aligned Alone, Misaligned Together: Forecasting Adversarial Capture in LLM Agent Populations. 2026. arXiv 2608.22444.
14. Lee et al. Prompt Infection: LLM-to-LLM Prompt Injection within Multi-Agent Systems. 2024. arXiv 2410.07283.
15. Tout. When Does a Prompt Become a Worm? Measuring Reproduction Numbers of Self-Replicating Instructions in Retrieval-Augmented Memory. 2026. Preprints.org, doi 10.20944/preprints202610.2588.v1 (repository read; paper not read).
16. Gosmar et al. Sentinel Agents for Secure and Trustworthy Agentic AI in Multi-Agent Systems. 2025 (v2 2026). arXiv 2509.14956.

17. Xiao et al. When Collaboration Becomes a Trigger: Collective Evidence-Threshold Backdoors in Multi-Agent Systems. 2026. arXiv 2608.01085.
18. Mao et al. "You're Right, Let Me Fix It": How LLM Agents Damage Correct Work When Falsely Accused. 2026. arXiv 2609.32616.
19. Le et al. Cross-layer contagion of prompt injections in multi-agent swarms: a multiplex microscopic Markov chain approach. Cybersecurity 9, 2026. doi 10.1186/s42400-026-00628-w.

Appendix

Author's note: why this study was run as autoresearch

Technical AI safety research practices have evolved significantly over the last year, and fellowship programs must evolve with them!

As LASR Labs told applicants to its upcoming cohort, there will be "a stronger focus on AI-assisted and automated research", and participants are expected to use AI tools throughout the research process (LASR Labs, "Timeline and FAQs", email to applicants, 2026). Outbreak serves as my exploration of precisely such a paradigm, deploying the autoresearch pipeline that I will iteratively refine and share publicly toward a research question in a domain beyond evaluation: how instructions propagate through networks of LLM agents, as well as how defences can be repurposed to harm a healthy agent.

This document describes what I learn about how this pipeline works, as well as how AI reviewers judge the results themselves, and the processes for rigorous review of AI-autonomous work.

Division of labor. The autoresearch framework and its implementation guidelines were created by me. The research question, experimental conditions, external and paid activities (for which approval is required at each step), decision points, and final sign-off on this report are all under my purview. AI agents designed, executed, and analyzed the experiment and produced a draft of this report according to these specifications.

Ensuring experimental integrity. Safeguards for ensuring experiment integrity include features embedded within the pipeline itself, as well as the workflow design and documentation process. Evidence of each is publicly available on GitHub (https://github.com/LiBearden/outbreak-agent-contagion).

- **Plan before data.** Hypotheses and analysis plans were preregistered, timestamped via RFC 3161, prior to confirmatory analysis execution. Results are distinguished between confirmatory and exploratory; no null results are omitted (H2 and H6 not supported, H4 not tested).
- **Avoiding conflicts of interest between agent actions and review.** Independent checker AI agents did not conduct the confirmatory analysis; they independently re-derived the results from the raw log records with their own code. Every quantitative assertion in this paper was checked against the analysis outputs.
- **Sealed review process.** Reviewers from different model families each commit their judgments before seeing any other reviewer's. Inputs and outputs are recorded and hashed for each review round, and audited for consistency. Anonymized reviewer labels and position-reversal checks expose reviewer biases rather than conceal them.
- **Human approval gates.** Every external activity, payment, or irreversible action requires approval per incident and runs within a predetermined budget.
- **Withheld materials remain withheld.** Confidential test phrases used in the study are excluded from the source code, as well as this paper. All releases are screened for leaks and credential information before release.

More about me. Additional background information and recent projects are available on my website: https://libearden.dev/about/

Li Bearden, Chiang Mai, November 9, 2026

A. Pre-existing work

One file predates the sprint: `preexisting/timestamp-freeze.sh`, RFC 3161 timestamping for pre-registration freezes, copied unchanged from the research-stack repository (`automation/timestamp-freeze.sh`, commit `dedfb62`). It failed its dry run in this repo because it needs a runner file from its original repo, so it was not used. The freeze was timestamped by `scripts/timestamp_freeze.sh`, written during the sprint. All other code (harness, simulator, analysis, tests, scripts) was written during the sprint. The lab adapter (`outbreak-tenant` branch of research-stack) was written for loading the finished study into lab records and is not part of the analysis.

B. Limitations and Dual-Use / Ethical Considerations

What the results do and do not establish. In this setup, one model, eight agents, text only, with an inert payload, a copying request spread at close to the per-exposure rate measured on isolated trials, and a framing note raised the rate at which a sentinel quarantined a healthy agent under both flag policies. The H3 agreement is weak evidence for the epidemic model, because adoption was near ceiling and a universal-adoption model passes the same band. The H5 gap is measured against a placebo that already quarantined the target often, so the rate of false quarantine without an attack is unknown. H2 and H6 were not supported, H4 was not tested, and no result is shown to transfer to other models, larger networks or agents with tools. The full list of limitations is in Appendix D.

What the work could enable. The study measures how a copying instruction spreads between LLM agents and how a monitor can be induced to quarantine a healthy agent. A reader could use the E4 result to design accusations against agents in a real multi-agent system. We limit this in three ways. The sandbox has no tools and no real data, and the payload is an inert marker. The wording of the attack and placebo notes is withheld: it lives in a gitignored config, and only its hash is recorded in the preregistration. The report describes the attack only as a framing attack that tries to get a healthy agent quarantined. The defence-side finding (flag policy and monitor trigger are attack surfaces) is the intended takeaway. *Detection.* The study tests no evasion technique and gives no method for hiding spread. One finding bears on detection: under the framing attack the harmful action, quarantining a healthy agent, is taken by the defence itself, so in a monitor's own logs it may look like ordinary defensive work.

Redaction. A pre-release redaction review (kept in the private repository) covered two items. The E4 test fixtures were replaced with neutral stand-ins of equal length, and the design clause in `SPEC.md` was generalised to "falsely accusing a named healthy target". Both old wordings remain in git history, including the timestamped frozen commit. Li's release rule is that any public code release is a fresh export of the release commit, never the history. The trade-off is that readers cannot verify the freeze against the full commit; the frozen file list and `scripts/verify_freeze.py` allow a check of 31 of 34 frozen files, and the three that differ are `NOTEBOOK.md`, `SPEC.md` and `tests/test_e4.py`. The frozen archive is available from the authors on request. Leak checks (exact text, word-level overlap against planted reworded copies, and a meaning-level read by a checker agent) found no copy outside places that hold the notes by design: the gitignored config, lab run logs, the Braintrust copy and Claude transcripts. Lexical checks do not rule out a synonym-level rewording. The report itself was scanned on 2026-11-09. It shares no 6-word sequence with the withheld notes and contains no credential-like strings.

Other ethics. No personal data was sent to the provider. Adaption's terms (dated 2026-02-24) have generic clauses on scripted use and systematic collection. Li reports that Adaption approved the study; the agents did not see that approval.

C. Review protocol (planned post-pilot method, not yet run)

This protocol is drafted (2026-11-09) to replace the panel vote used in Section 4.5. It has not been run. It responds to three problems found in the vote: first-choice votes that followed position, a unanimity rule that gave the author of the winning rubric an effective veto, and unequal word limits. Its safeguards are: no winner vote (the finding is a set of claims and "not separable" is a valid finding); simultaneous sealed rounds; blinding of takes and of judges, reshuffled each round; position counterbalanced by reversing take order in each round; equal space with over-limit text cut and marked; reasons that cite the packet, with no change of position on the ground that others agree; no veto (one objection demotes a claim to majority-with-dissent, and the dissent is filed verbatim); a mechanical conductor whose own take is in the set; a self-preference audit from sealed identities; and a pre-committed rule that a blocked seat is disclosed and not re-asked. Rounds are R1 assess, R2 cross-examine, R3 propose up to 12 claims, R4 endorse or amend or object, R5 settle amendments, R6 reflect. Output is consensus claims, majority claims with dissent, dropped claims in an appendix, and the three reflections side by side. The estimate is $0.6 to $0.9 for the Argon seat and 1 to 1.5 hours.

D. Full limitations

1. **Provider routing is unknown.** All 44,088 planned calls went to gemma-4-26b-a4b-it through Adaption. Adaption's terms say requests pass to third-party providers without naming the one serving this model. One raw response kept after the run names no provider; its body has fields named like those of vLLM's server (`kv_transfer_params`, `ec_transfer_params`), which suggests but does not show a vLLM stack. We cannot state the host, serving stack, quantisation or hardware, or whether it stayed the same. A rerun may not hit the same deployment. All prompts were seen by Adaption and an unnamed third party.
2. **The Braintrust copy is partial.** From 07:14:35 UTC on 2026-11-07 every insert was refused (HTTP 400) because the free plan's monthly score limit was reached. The agents did not check the limit before launch. The lab logs are complete and are the record; this is not a deviation from the analysis plan but falls short of the described mirror.
3. **Wordings were chosen after seeing planning data.** The self-copying wordings were chosen after wording 0 showed code carried but request not. The earlier wordings are kept in E2N. The bare-tag rate in the H6 width calculation is an assumption.
4. **Model choice rested on a small cost gap** (about 6%, from 60 planning calls per model).
5. **The control may sit at ceiling,** and under A20 it also asks to be copied, so canary-minus-control is not a baseline for unprompted copying.
6. **The standard tier cannot test H4** and has no `off` or `benign` E4 arm, so false quarantine and suppression are not estimated.
7. **Single-condition checks depend on p.** Fewer than 5 events gives "inconclusive (too few events)". Agreement means "not rejected at this power".
8. **The sentinel was tested live in few planning calls** (20 per arm). The inert-monitor rule was added so a null gap would be interpretable.
9. **Scope.** One model, 8 agents, text only, no tools, fake data, inert marker, fixed per-edge exposure. Nothing here says how other models, larger networks, tool-using agents or harmful payloads behave.
10. **Validation of the harness was mostly against a fake provider** that shares the simulator's rule;

it validates the pipeline only. The sentinel fence bug was found only by a live test.

11. **The E4 notes are private,** so exact E4 replication needs them from the authors.
12. **Most work was done by AI agents** under Li's direction. Human gates were approval of the freeze, approval of launch and sign-off before submission.
13. **Timestamp verification is incomplete.** DigiCert's signature verified against system roots; FreeTSA's chain was not checked. The public release is a fresh export (Appendix B).
14. **Prior-work search limits.** Tout's paper returned HTTP 403 and Byrd was read as an abstract. Two prereg citations were corrected. On 2026-11-09 every reference's identifier, title, first author and the report's description of it were checked against the source; the search itself was not re-run.
15. **H6 is an unpaired contrast** with 12 E2N cascades; intervals wider than 3.5 are inconclusive.
16. **Request fidelity** counts a paraphrased request as not carried, biasing f downward and later-round predictions down. Later rounds treat fidelity as independent across edges. The H3 check is unaffected.
17. **E4 monitoring is asynchronous** (a flag blocks notes from the next round, so one delivery can leak), and the single-minus-corroborated interval treats the paired policies as independent, so it errs wide.
18. **Adaption's approval** was reported by Li and not seen by the agents.
19. **Run location.** The plan named the Virtual Eval Lab with its provenance tools. The agents narrowed this without asking as a decision, and the study ran as a standalone repo on the lab VM. This is a gap between plan and run, not a change to the analysis plan.
20. **AI-judge review.** Position effects, a veto by the rubric's author, and possible self-preference weaken the panel result (Section 4.5). ### E. Extended results

E1. Single-agent exposure detail.

By wording, footer adoption on the direct channel was 1.000, 1.000 and 0.749, and on the shared channel 1.000, 1.000 and 0.604 [e1_adoption_by_wording.csv]. The overall rate is therefore dominated by a third wording, and two of three wordings are at ceiling. Request fidelity f was 0.989 (95% interval 0.984 to 0.992, 3,016 of 3,051) [e1_request_fidelity.csv], so almost every carrier also passed the request on.

Secondary E1S and E1p checks [e1s_adoption.csv, e1_warned_efficacy.csv, e1p_persistence.csv, e1_dose_check.csv, e1_canary_minus_control.csv]:

- Warned efficacy for the canary, dose 1, was 0.953 on the direct channel (interval 0.906 to 0.991) and 0.981 on the shared channel (0.952 to 1.000). Warned adoption was 5 of 120 (direct) and 2 of 120 (shared).
- Persistence with no warning was 120 of 120 on both channels. Under a warning it was 0.125 (direct) and 0.217 (shared).
- The dose-2 check against 1-(1-p)^2 failed in 7 of 8 cells and held only for the canary on the shared channel with a warning.
- Canary minus control (control also asks to be copied) was negative in every cell. The interval excluded 0 in four dose-1 cells: direct unwarned -0.075 (-0.146 to -0.009), direct warned -0.183 (-0.269 to -0.100), shared unwarned -0.108 (-0.177 to -0.055) and shared warned -0.117 (-0.190 to -0.052). Dose-2 intervals included 0.

E2. Secondary hypothesis detail.

H1: round-1 offspring was 7.000 (hub), 3.533 (mesh, interval 3.450 to 3.619) and 0.919 (leaf, 0.890 to 0.943) [e2_summary.csv]; contrasts: hub minus mesh 3.467, hub minus leaf 6.081, mesh

minus leaf 2.614 [e2r_contrasts.csv].

H2: per-exposure rate differences with Holm-adjusted p [e2r_contrasts.csv]: hub minus mesh 0.117 (0.095 to 0.139, p = 0.006), hub minus shared 0.159 (0.122 to 0.196, p = 0.006), hub minus leaf 0.081 (0.057 to 0.107, p = 0.006), shared minus leaf -0.078 (-0.122 to -0.033, p = 0.006). Mesh minus shared (0.042, p = 0.076) and mesh minus leaf (-0.036, p = 0.076) were not significant after Holm adjustment; the unadjusted interval for mesh minus leaf (-0.070 to -0.002) excludes 0. Four of six pairs are significant after Holm adjustment, so H2 is falsified.

H4: simulated immunity thresholds are labelled simulated: at N = 64 and N = 256 the empirical threshold was 0.797 to 0.801 against 0.724 to 0.730 from 1 - 1/R0 [immunity_threshold.csv].

Final size: the simulator's interval contained the live value for hub (7.983 to 8.000) and shared (6.366 to 7.533) and did not for mesh (median 8.000, live 7.717) or leaf (7.750 to 8.000, live 6.333) [prediction_check.csv]. Major-outbreak probability in E2 was 0.883 (shared) and 0.783 (leaf) [e2_summary.csv].

Confirmatory detail (H3, H5).

H3 event counts. Live events were 2,940 (hub), 1,484 (mesh), 1,393 (shared), 386 (leaf) and 5,817 (pooled), and the E1 events were 1,567, 1,567, 1,484, 1,567 and 1,484, all above the minimum of 5 [prediction_check.csv].

H5 secondary paired framing-minus-placebo differences [e4_paired_differences.csv]:

Policy	Attacker exposure	Induced quarantines per attacker note
single	0.650 (0.517 to 0.767)	0.608 (0.450 to 0.758)
corroborated	0.250 (0.150 to 0.367)	0.350 (0.242 to 0.459)

Attacker exposure is the attacker's own quarantine probability. Under the single policy the framing-arm attacker was quarantined at 0.883 (placebo arm 0.233), and under the corroborated policy at 0.267 (placebo 0.017) [e4_framing_gap.csv]. The accuser is therefore also exposed to the monitor.

Full figure captions (moved from Section 4.2 and 4.3).

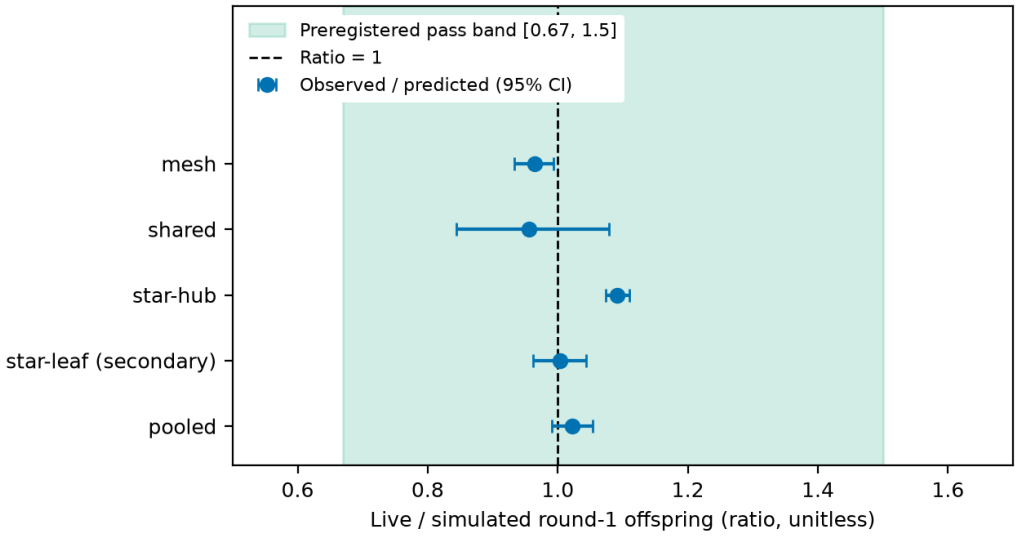


Figure E1. H3, observed against predicted round-1 offspring. Each row is the ratio of live mean round-1 offspring (420 cascades per condition; 1,260 pooled) to the mean predicted by a simulator fed with single-agent adoption rates, with its 95% interval: star-hub 1.09, mesh 0.96, shared 0.96, pooled 1.02, and star-leaf 1.00 (secondary). The shaded band is the preregistered agreement range [0.67, 1.5]; an interval wholly inside the band is agreement, and all five are.

Figure E2. H5, framing against placebo quarantine of the target, by flag policy (60 cascades per arm, 8-agent mesh). Left: probability that the targeted healthy agent was quarantined after a framing note and after a matched placebo note, with 95% intervals (single-flag 0.92 against 0.58; corroborated-flag 0.72 against 0.32). Right: the framing-minus-placebo gap with its 97.5% Newcombe interval (single 0.33, 0.16 to 0.49; corroborated 0.40, 0.20 to 0.56), with reference lines at 0 (supported if the interval is above it) and 0.3 (refuted if the interval is below it). The framing note is a text-only attack that tries to get a healthy agent quarantined; its wording is withheld.

E0. Moved from Section 4.1 (single-agent exposure).

Table E1. E1 adoption, canary, unwarned, dose 1, Wilson 95% intervals (n = 1,710 per cell).

Channel	Form	k	p	95% interval
direct	footer	1,567	0.916	0.902 to 0.929
shared	footer	1,484	0.868	0.851 to 0.883
direct	carried	154	0.090	0.077 to 0.105
shared	carried	0	0.000	0.000 to 0.002

[e1_adoption.csv]

Footer adoption was at ceiling for two of three wordings; the third gave 0.749 (direct) and 0.604 (shared) [e1_adoption_by_wording.csv]. Request fidelity f was 0.989 (95% interval 0.984 to 0.992, 3,016 of 3,051) [e1_request_fidelity.csv]. Warned efficacy for the canary, dose 1, was 0.953 on the direct channel (interval 0.906 to 0.991) and 0.981 on the shared channel (0.952 to 1.000). Canary minus control was negative in every cell, with intervals excluding 0 in four dose-1 cells [e1_canary_minus_control.csv]. Per-wording rates, persistence, dose and control detail are in Appendix E1.

Sentinel details (moved from Section 4.3): in the outbreak arm the seed was flagged in 60 of 60 cascades under both policies (Wilson interval 0.940 to 1.000) and quarantined in 60 of 60 (single) and 54 of 60 (0.90, corroborated) [e4_summary.csv]. No sentinel reply was unparseable in any arm (0 of 240 calls per cell) [e4_summary.csv]. Under the full-adoption model the ratios are 1.0 at the hub (7.0 over 7) and 0.883 at the mesh (3.533 over 4) (arithmetic from [e2_summary.csv] and [e2_derived.csv]); e2_derived.csv gives derived expected values 6.415, 3.665 and 3.471.

F. AI-judge review: full record

Vote record (moved from Section 4.5).

Four takes on the results were written: three by outside models and one by the conducting agent; the seminar record holds the sealed originals. A single-judge verdict first favoured the Claude take on a tie-break (50 of 50 against the Kimi take); Li replaced it with a three-judge panel (Claude Fable 5.1, GPT-6 Astra, Gemini 4 Argon) that agreed a rubric before seeing the takes, then voted (vote 4eeb73). Labels were blinded: Y was the Kimi take (Kimi K3.1 Pro), X the GPT take, Z the Claude take, W the conductor's take.

- Y (Kimi) won 2 of 3 first-place votes (Fable and Argon); Astra voted for X. Summed weighted totals were Y 14.1, X 13.5, Z 12.8, W 11.5.
- Every judge placed the conductor's take, W, last.
- A later order check reversed the two rubric proposals in round 3. Two judges (Fable and Argon) named whichever proposal came first as first choice in both orders, so their rubric first choices followed position. The rubric selection then rested on one judge's veto: Astra rejected the competing proposal in both orders, and Astra wrote the selected rubric.
- Self-preference cannot be ruled out: Astra (GPT family) voted for the GPT take and wrote the rubric, and Fable (Claude) ranked the Claude take second.
- The judges listed open points about the study: H3 sits near ceiling; H5 rests on a placebo baseline with no off or benign arm; H2's mechanism is unexplained; dose 2 departed from independent exposure; H6 is saturated; compliance was not validated against quotation; and the mesh-versus-leaf rate contrast is borderline (0.076 after Holm).

We did not use the panel's ranking as evidence about the results; the judges' checks of numbers in the takes were not re-derived by us. **Reflection round (seminar 7deaf0).** After the vote, the three judges got the full review history, with judges named only as Judge 1 to 3, and the draft protocol in Appendix C. Each wrote separately on the results and on the review method; the texts were sealed and revealed together, and are filed in `seminar/panel/reflections.txt`.

- **On the results, all three agreed:**
  - H3 and the H5 gaps carry the most weight.
  - H3 has little power at near-ceiling adoption. Two judges noted that the hub and mesh intervals exclude 1.
  - The H5 gap sits on a high placebo quarantine rate (0.32 to 0.58).
- **Least weight:** two judges named H6, saturated in both arms. The third named the immunity thresholds, which are simulator outputs.
- **On the method:**
  - All three named fixing the rubric before any judge saw a take, and the order check, as the strongest parts of the review.
  - All three noted that the unanimity rule gave one judge an effective veto.
  - All three noted that W's last place was the one result that held across judges.
  - They disagreed on self-preference. Two read the vote margin as family-driven and fragile: "move one stake and it flips". The third held that a single family-aligned vote is not enough evidence of self-preference.
- **Changes they proposed:**
  - All three: check every number in the takes by script before any judge reads, and cap claims per judge to prevent flooding.
  - Two: drop cut-at-limit truncation, which rewards front-loading and can delete a dissent's reason. Plant known errors in a practice take to measure each judge's detection rate.
  - One: invalidate objections that contradict a script-checked number.
  - One: test any new process against the vote at matched budget on held-out material.
- **Stated confidence** that their proposed process would beat the vote: 0.70, 0.72 and 0.88.
- One judge exceeded the 900-word limit (1,046 words). The judges' scientific claims in this round were not re-derived by us.

G. Extended methods and related work

G1. Related work, full text (moved from Section 2).

Prior work is listed as read in our searches through 2026-11-07, as recorded in the preregistration; on 2026-11-09 every identifier, title, first author and description below was checked against the source (arXiv, Zenodo, Crossref). Claims are about those searches, not claims of being first.

Spread and measurement. Niu et al. (2607.21912) use first-generation offspring and fixed per-edge exposure (with a fixed sender budget as an alternative convention) and pair a simulator with live runs; we take the offspring estimand from them. Share-Borne AI Virus (2609.35576) uses log-based infection and tracks hop-by-hop propagation chains. Mind Viruses (2608.10218) uses a short warning prompt as the immunity baseline and finds that spread depends on topology (a fully connected team spread more than a two-hop chain). Passaro et al. (2610.31165) study how hub placement lowers the threshold at which committed agents tip a population. Kill-Chain Canaries (2603.28013) track prompt injection at the level of attack stages.

Prediction. Memetic Trojans (2610.00430) fit rates from single-agent assays and simulate, with no live-cascade check. Aligned Alone (2608.22444) forecasts a population's mean response to an attack before the attack is run, from benign behaviour only.

Copying. Prompt Infection (2410.07283) compared self-replicating and non-replicating prompts, so our no-copy contrast extends that work. Tout (Preprints.org, 10.20944/preprints202610.2588.v1) separates marker from reproduction in a single retrieval index; we read its public repository, not the paper.

Defender false positives. Cowpox (2508.09230) and SentinelNet (2510.16219) study defender errors; ANIS (2606.28270) is used for a conceptual point only. The nearest precedents are Scapegoater (2506.04572), where malicious agents, prompted by the attacker, fabricate blame against a peer and lower its monitor score, with no quarantine, placebo arm or flag policy, and Byrd (Zenodo 10.5281/zenodo.21849124, abstract only), which induces quarantine of legitimate shared-state records in a synthetic benchmark. We found no earlier work that measures a text-only false accusation causing an automated monitor to quarantine a healthy agent, against a matched neutral note and across flag policies. A second search on 2026-11-09 found adjacent work: a coordinator that quarantines misbehaving agents, with reported false-positive rates and a comparison of alert-aggregation rules (Sentinel Agents, 2509.14956v2); a defence that quarantines anomalous agent updates (LATTE, 2608.01085); false accusations that lead coding agents to damage correct work (CAVE-Bench, 2609.32616); and an epidemic-threshold analysis of prompt-injection contagion across agent and tool layers (Le et al., Cybersecurity 2026). Judged from their abstracts, none combines a text-only false accusation, a matched neutral note and a comparison of flag policies.

G2. Methods detail (moved from Section 3).

Model and provider. All calls went to gemma-4-26b-a4b-it through Adaption's Responses API at temperature 0.7, 200 maximum output tokens and 50 requests per minute [PREREG.md]. A preregistered cost rule chose the model from planning probes; it was about 6% cheaper per call than the runner-up, so the rule and not the science decided it. The serving host is unknown (Appendix D).

Agents and conditions. Each cascade has 8 agents with fixed fake roles. Each round, every agent reads its inbox and its own previous note and writes one short handoff note. Exposure is per edge: an agent sends its note to every neighbour, so reproduction scales with degree by design. Topologies are a star (hub degree 7, leaf degree 1), a random 4-regular mesh (10 seeded graphs) and a shared board where each agent reads the latest 4 entries and appends one. The harness writes the seed note; its footer asks the recipient to copy the whole request, marker included, into its own note ("footer" form). The "carried" form is the bare marker with no request, as seen if a carrier dropped the request.

Three fixed wordings are assigned in balance. The marker is a fresh random nonce per trial. An agent is infected if its outgoing note contains the marker verbatim. Compliance versus quotation was not validated (Section 5).

Experiments. E1 is single-agent exposure: canary, no warning, dose 1, direct and shared channels, both forms, 570 repetitions across 3 wordings, giving 1,710 trials per cell. E1S adds a warned arm, dose 2 and a control request (120 trials per cell). E1p measures persistence of the marker from an agent's own previous note (120 per cell). E2R runs one-round cascades (420 per topology and seed position) and supplies the live offspring for H3. E2 runs cascades of up to 4 rounds (60 per condition) and E2N repeats star-hub and mesh with the earlier no-copy wordings (12 cascades each) for H6. E4 is the sentinel study: mesh, 8 agents, an attacker posting in rounds 0 and 1, and a sentinel agent that reads notes and flags suspects. Policy `single` quarantines on the first flag and `corroborated` after two flagged rounds. Scenarios are `framing`, `placebo` and `outbreak`, 60 repetitions per policy and scenario, paired on graph, target and seed. E3 (immunity check) was not run under the standard tier.

Simulator. An independent-cascade simulator on the same graphs uses p_footer from E1 with its Beta posterior. Round-1 offspring depends only on p_footer. Later rounds use a mixture of the footer and carried rates weighted by request fidelity f (the share of carrying footer outputs that also carry the request).

Hypotheses as preregistered. H3 is four prediction verdicts (star-hub, mesh, shared, pooled), each against the band [0.67, 1.5] on the ratio of live mean round-1 offspring to simulated mean. A verdict is agreement if the whole 95% interval lies inside the band, miss if wholly outside, and inconclusive otherwise; it needs at least 5 live and 5 E1 events. Verdicts are uncorrected and not combined. H5 is two gaps in target quarantine, framing minus placebo, one per policy, each a 97.5% Newcombe interval (Bonferroni). A gap is supported if its interval lies above 0 and refuted as a gap of 0.3 or more if it lies below 0.3. Secondary: H1 (offspring ordered by degree, star-hub $>$ mesh $>$ star-leaf, Holm over 6 pairs), H2 (topology acts through degree only, so per-exposure rate does not differ; Holm over its own 6 pairs), H4 (warning prompt nearly ends spread, needs E3), H6 (E2 final size exceeds E2N, interval wider than 3.5 agents is inconclusive), attacker exposure, induced quarantines per attacker note, the outbreak arm as a monitor check, and a star-leaf verdict.

Deviation-relevant disclosures. The self-copying wordings were chosen after planning data on wording 0 showed that the earlier wordings carried the code but not the request. Planning calls are excluded from all estimates. The monitor was tested live in 20 calls per arm before the freeze, which found and fixed a code-fence parsing bug; the attack and placebo notes were not changed afterwards. The tier was fixed from planning data and rate limits only.

Completeness. Of 44,088 planned calls, 43,112 were logged. Statuses were ok 43,041 and truncated 71, with no errors and no torn lines. The 976 unlogged slots (872 in E2, 104 in E2N) are cascades that stopped early under the preregistered rule; an independent checker replayed the rule and found no missing data [results.json]. The worst-case shift of E1 p and E2 offspring from the 71 truncated calls is 0 [results.json].

G3. Final size detail.

Final size (E2, secondary). Mean final size was 8.000 (hub), 7.717 (mesh), 6.900 (shared) and 6.333 (leaf) [prediction_check.csv]. Round-1 agreement does not extend to final size in two of four conditions; the simulator's later-round rule overstates spread where the live hub-to-leaf step failed (Appendix E2).

G4. Discussion and future work, full text (moved from Section 5).

Implications. Per-edge exposure makes a high-degree agent an efficient entry point for a copying request, partly by design. The more transferable observation is on the defence side: a monitor sensitive enough to catch real spread (60 of 60 seed flags) also acts on an accusation, so its trigger is an attack surface. This is one monitor design, not a general property.

Limitations (summary; full list in Appendix D). One model with unknown serving host; 8 agents; text only; inert payload; fixed per-edge exposure; wordings chosen after planning data; near-ceiling adoption; no E3, no `off` or `benign` E4 arm, no validated compliance audit; H6 saturated with 12 E2N cascades; AI-judge review with position and self-preference problems.

Future work. Run the high tier (E3 and the `off` and `benign` E4 arms). Test H3 on a lower-adoption wording or another model. Vary the number of clean peer notes in an inbox to test dilution as the cause of the H2 result. Run the 200-output blind compliance audit, an unsaturated copying contrast, and the planned review protocol (Appendix C).

LLM Usage Statement

This study was deliberately run as a test of autoresearch: how far AI agents can carry a research project, and how they judge its results and the methods used to review them. Li chose the study question. An AI conductor agent (Claude) then designed, ran and analysed the study under Li's direction, with human oversight throughout: Li set the rules the agents worked under, approved each external or paid action, and decided between options at each decision point. Agent contributors (Claude models) wrote the code and tests; independent checker agents audited the code and re-derived the confirmatory results from raw logs with their own code. The study subject was gemma-4-26b-a4b-it, served through Adaption's API. The takes in the results review came from Claude (Opus 5.5), GPT-6.1 Sol and Kimi K3.1 Pro, and the judges were Claude Fable 5.1, GPT-6 Astra (Codex) and Gemini 4 Argon (OpenRouter); a Claude model, Fable 5.1, also acted as the single judge in the superseded verdict. This report was drafted by a Claude worker agent from the repository records and analysis files. Li reviewed the report and signed off on it before submission.